

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Randomized, Multicenter, Open-Label, Phase III Clinical Trial to Evaluate the Efficacy, Safety, and Pharmacokinetics of Prophylactic Emicizumab Versus No Prophylaxis in Hemophilia A Patients
Short Title:	Efficacy, Safety, and Pharmacokinetic Study of Prophylactic Emicizumab Versus No Prophylaxis in Hemophilia A Participants
Protocol Number:	622471783553308421
NCT Number:	NCT03315455
Sponsor Name:	Roche
Investigational Product:	emicizumab
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Model-Based Annualized Bleeding Rate for Treated Bleeds: The number of treated bleeds over the efficacy period was estimated as an annualized bleeding rate (ABR) using a negative binomial regression model, which accounts for different follow-up times. A treated bleed was defined as a bleed that was directly followed by a hemophilia medication reported to be a "treatment for bleed". The 72-hour rule was implemented: two bleeds of the same type and at the same anatomical location were counted as one bleed if the second bleed occurred within 72 hours from the last treatment for the first bleed. Bleeds due to surgery/procedure were excluded. 2. Mean Calculated Annualized Bleeding Rate for Treated Bleeds: The number of treated bleeds over the efficacy period is presented here as a calculated annualized bleeding rate (ABR) that was annualized for each participant using the following formula: $ABR = (\text{number of bleeds} / \text{number of days during the efficacy period}) \times 365.25$. A treated bleed was defined as a bleed that was directly followed by a hemophilia medication reported to be a "treatment for bleed". The 72-hour rule was implemented: two bleeds of the same type and at the same anatomical location were counted as one bleed if the second bleed occurred within 72 hours from the last treatment for the first bleed. Bleeds due to surgery/procedure were excluded. 3. Median Calculated Annualized Bleeding Rate for Treated Bleeds: The number of treated bleeds over the efficacy period is presented here as a calculated annualized bleeding rate (ABR) that was annualized for each participant using the following formula: $ABR = (\text{number of bleeds} / \text{number of days during the efficacy period}) \times 365.25$. A treated bleed was defined as a bleed that was directly followed by a hemophilia medication reported to be a "treatment for bleed". The 72-hour rule was implemented: two bleeds of the same type and at the same anatomical location were
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counted as one bleed if the second bleed occurred within 72 hours from the last treatment for the first bleed. Bleeds due to surgery/procedure were excluded.

Secondary Obj.:

1. Model-Based Annualized Bleeding Rate for All Bleeds
2. Mean Calculated Annualized Bleeding Rate for All Bleeds
3. Median Calculated Annualized Bleeding Rate for All Bleeds

2.2 Background & Rationale

This multicenter, open-label, Phase 3 study with randomized and non-randomized arms is designed to investigate the efficacy, safety, and pharmacokinetics of emicizumab in participants with hemophilia A regardless of factor VIII (FVIII) inhibitor status. Participants greater than or equal to (≥)12 years old who received episodic therapy with FVIII or bypassing agents prior to study entry and experienced at least 5 bleeds over the prior 24 weeks will be randomized in a 2:2:1 ratio to the following regimens: Arm A: Emicizumab prophylaxis at 3 milligrams per kilogram (mg/kg) once every week (QW) subcutaneously (SC) for 4 weeks, followed by 1.5 mg/kg QW SC; Arm B: Emicizumab prophylaxis at 3 mg/kg QW SC for 4 weeks, followed by 6 mg/kg once every 4 weeks (Q4W) SC; and Arm C: No prophylaxis (control arm). In addition, pediatric participants less than (<)12 years old with hemophilia A and FVIII inhibitors who received episodic therapy with bypassing agents prior to study entry will be enrolled to Arm D: Emicizumab prophylaxis at 3 mg/kg QW SC for 4 weeks, followed by 1.5 mg/kg QW SC.

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: COMPLETED

Target N: 85

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

Inclusion Criteria for Arms A, B, and C:

- * Diagnosis of severe congenital hemophilia A or hemophilia A with FVIII inhibitors
- * Aged 12 years or older at the time of informed consent
- * Body weight ≥40 kilograms (kg) at the time of screening
- * Participants without FVIII inhibitors (<0.6 Bethesda unit per milliliter [BU/mL]) who completed successful immune tolerance induction (ITI) must have done so at least 5 years before screening and have no evidence of inhibitor recurrence (permanent or temporary)
- * Documentation of the details of episodic therapy (FVIII or bypassing agents) and of number of bleeding episodes for at least the last 24 weeks and ≥5 bleeds in the last 24 weeks prior to study entry
- * Adequate hematologic, hepatic, and renal function
- * For women of child bearing potential: agreement to remain abstinent or use a protocol defined contraceptive measure during the treatment period and for at least 5 elimination half-lives (24 weeks) after the last dose of study drug

Inclusion Criteria for Arm D:

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- * Diagnosis of congenital hemophilia A of any severity and documented history of high-titer inhibitor (i.e., ≥ 5 BU/mL)
- * Children ≤ 12 years old at time of informed consent
- * Body weight ≥ 3 kg at time of informed consent
- * Requires treatment with bypassing agents
- * Adequate hematologic, hepatic, and renal function
- * For female participants who are of childbearing potential, follow the same contraception criteria as listed above for Arms A, B, and C

4.2 Exclusion Criteria

Exclusion Criteria for Arms A, B, and C:

- * Inherited or acquired bleeding disorder other than hemophilia A
- * At high risk for thrombotic microangiopathy, in the investigator's judgment
- * History of illicit drug or alcohol abuse within 48 weeks prior to screening, in the investigator's judgment
- * Previous (in the past 12 months) or current treatment for thromboembolic disease (with the exception of previous catheter-associated thrombosis for which anti-thrombotic treatment is not currently ongoing) or signs of thromboembolic disease
- * Other conditions that may increase risk of bleeding or thrombosis
- * History of clinically significant hypersensitivity associated with monoclonal antibody therapies or components of the emicizumab injection
- * Known human immuno-deficiency virus (HIV) infection with cluster of differentiation 4 (CD4) count < 200 cells/microliter (cells/mL) within 24 weeks prior to screening. Participants with HIV infection who have CD4 ≥ 200 cells/mL and meet all other criteria are eligible
- * Use of systemic immunomodulators at enrollment or planned use during the study, with the exception of anti-retroviral therapy
- * Concurrent disease, treatment, or abnormality in clinical laboratory tests that could interfere with the conduct of the study, may pose additional risk, or would, in the opinion of the investigator, preclude the participant's safe participation in and completion of the study
- * Planned surgery (excluding minor procedures such as tooth extraction or incision and drainage) during the study
- * Receipt of: Emicizumab in a prior investigational study; An investigational drug to treat or reduce the risk of hemophilic bleeds within 5 half-lives of last drug administration; A non-hemophilia-related investigational drug concurrently, within last 30 days or 5 half-lives, whichever is shorter
- * Pregnant or lactating, or intending to become pregnant during the study

Exclusion Criteria for Arm D:

- * Inherited or acquired bleeding disorder other than hemophilia A
- * Ongoing (or plan to receive during the study) ITI therapy or prophylaxis treatment with FVIII
- * Previous (in the past 12 months) or current treatment for thromboembolic disease (with the exception of previous catheter-associated thrombosis for which anti-thrombotic treatment is not currently ongoing) or signs of thromboembolic disease

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- * Other diseases that may increase risk of bleeding or thrombosis
- * History of clinically significant hypersensitivity associated with monoclonal antibody therapies or components of the emicizumab injection
- * Known infection with HIV, hepatitis B virus (HBV), or hepatitis C virus (HCV)
- * At high risk for thrombotic microangiopathy, in the investigator's judgment
- * Use of systemic immunomodulators at enrollment or planned use during the study
- * Planned surgery (excluding minor procedures such as tooth extraction or incision and drainage) during the study
- * Inability (or unwillingness by caregiver) to receive (allow receipt of) blood or blood products (or any standard-of-care treatment for a life-threatening condition)
- * Receipt of: Emicizumab in a prior investigational study; An investigational drug to treat or reduce the risk of hemophilic bleeds within 5 half-lives of last drug administration; A non-hemophilia-related investigational drug concurrently, within last 30 days or 5 half-lives, whichever is shorter
- * Concurrent disease, treatment, or abnormality in clinical laboratory tests that could interfere with the conduct of the study, may pose additional risk, or would, in the opinion of the investigator, preclude the participant's safe participation in and completion of the study
- * Pregnant or lactating, or intending to become pregnant during the study

11. Study Identifiers & Governance

Primary ID: 622471783553308421

Registry Number: NCT03315455

Sponsor: Roche

Study Title: Efficacy, Safety, and Pharmacokinetic Study of Prophylactic Emicizumab Versus No Prophylaxis in Hemophilia A Participants

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____